

Fig. 4: Front panel of the comparison of wav file VI

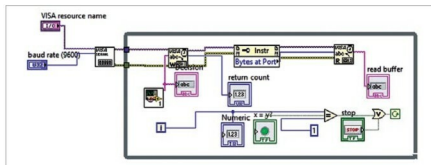


Fig. 5: Serial communication VI

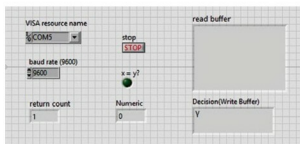


Fig. 6: Front panel of serial communication VI

in the system. With the help of a microphone, real-time sound signal is acquired by LabVIEW functional block. This acquired sound is not suitable for cross-correlation or comparison because of the noise within the real-time speech.

So it is advisable to filter the acquired signal using low-pass Butterworth filter to remove unwanted frequencies or noises. Low-pass Butterworth filter gives flat frequency response. Output of the filter is given to the cross-correlation block

or VI, shown in Fig. 2. Filter parameters are configured by setting low-pass Butterworth filter as cut-off frequency 3000Hz and filter order 4. Device ID is set as '0' for one-input device. Remaining block parameters are taken as by default.

Wave file creation VI creates or records wav file from audio command given by the user. The wav file is stored automatically on the default path. The path is shown on the label named 'Generated File Path' in the front panel of VI. The

to compare this signal with pre-stored command. We have used AT89S51 as controller.

The project includes the following programs: Wav file Creation Sub VI.vi, Cross Correlation.vi, Serial Communication.vi for front end LabVIEW program and Voice.c for the AT89S51 microcontroller.

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